In 1963 semi-dwarf wheat of Mexico was introduced from which India got five prolonged strategies for breeding a wide range of high varieties like Sonora 64, Sonalika and Kalyansona possessing a broad spectrum of resistance to major biotic and abiotic condition. Same as wheat M.S.Swaminathan produced the first semi-dwarf fertiliser responsive hybrid variety of rice_TNI (Taichung Native-1) in 1956 from Taiwan. The derivatives were introduced in 1966. Later better yielding semi dwarf varieties of rice Jaya and Ratna developed in India.



NORIN 10 – The cultivars found that Norin 10 dwarfing genes have high photosynthetic rate per unit leaf area and increase respiratory activity. Gonjiro Inazuka selected the semi-dwarf wheat variety that became Norin 10. He would have never thought that the semi dwarf genes would not only revolutionize the world of wheat but also helped to save more than one billion lives from hunger and starvation.

Plant Breeding for Developing Resistance to diseases

Some crop varieties bred by hybridization and selection, for disease resistance to fungi, bacteria and viral diseases are released (Table 9.1).

Crop	Variety	Resistance to diseases
Wheat	Himgiri	Leaf and Stripe rust, hill bunt
Brassica	Pusa swarnim (Kara rai)	White rust
Cauliflower	Pusa Shubhra, Pusa snowball K-1	Black rot and curl blight black rot
Cowpea	Pusa Komal	Bacterial blight
Chilli	Pusa Sadabahar	Chilly mosaic virus, Tobacco mosaic virus and Leaf curl.

Table 9.1: Disease resistance varieties

Norman E. Borlaug: The plant pathologist plant breeder devoted his life at the International Maize and Wheat improvement centre at Sonord in Mexico. He developed a new high yielding, rust resistant, non-lodging dwarf wheat varieties like Norin-10, Sonora-64, Lerma rojo-64, etc. which are now being cultivated in many countries. This formed the base for 'green revolution'. He was awarded a Nobel prize for Peace in 1970.



Dr. M. S. Swaminathan: He is a pioneer mutation breeder. He has produced Sharbati Sonora, is the amber grain coloured variety of wheat by mutation, which is responsible for green revolution in India.



Dr. Swaminathan is called "Father of green revolution in India".

Nel Jayaraman: Mr. Jayaraman, hails from Adirangam village in Tiruvarur district. He was a disciple of Dr.Nammalvar and state co-ordinator of 'Save our rice campaign, Tamil Nadu. He strived hard for conservation of traditional rice varieties. He had trained a team of farmers and regularly update them on the current issues that affect them.



In 2005, he organized a first ever traditional paddy seed festival in his farm as an individual. The seed festival in May 2016 at Adhirangam was 10th in a row and in which 156 different traditional varieties were distributed to more than 7000 farmers across Tamil Nadu. He was invited by the Philippines Government to give a talk at the International Rice Research Institute (IRRI) on his work and mission. In 2011, he received the State Award for best organic farmer for his contribution to organic farming, and in the year 2015, he received the National Award for best Genome Savior.

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